Type MS · Name the parts of a nucleotide and the DNA backbone

Answer 1 of 7 · Question 1 part c · 3 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

1 (a) (i)	**D** sugars A/B/C are incorrect as amino acids/bases/protein do not form part of backbone		1
(ii)	**B** bases A is incorrect as the structures labelled Y are bases C is incorrect as proteins are not incorporated into DNA structure D is incorrect as sugar is found in the backbone		1
(iii)	Any two from the following: • hydrogen bonds; • between complementary bases;	Allow description of complementary base pairing	2
(iv)	nucleus;		1
(b) (i)	Any two from the following: • RNA contains ribose sugar and DNA contains deoxyribose; • RNA is single-stranded and DNA is double stranded; • RNA contains uracil and DNA contains thymine;		2
(ii)	**D** mRNA and tRNA A/B/C are incorrect as pRNA/dRNA/dRNA are not types of RNA		1
(c)	**R** phosphate; **S** sugar/deoxyribose; **T** base;	Allow a named base	3
			Total 11

Type MS · Name the parts of a nucleotide and the DNA backbone

Answer 1 of 7 · Question 1 part c · 3 marks

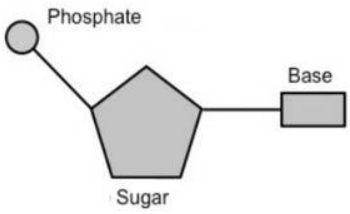
2022 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name the parts of a nucleotide and the DNA backbone

Answer 2 of 7 · Question 8 part a.v · 3 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a.v in this mark scheme extract

8 (a)	(i)	Any two from: • (movement of substances/named solute or gas) against a concentration gradient(1) • using energy/ATP(1)		2
	(ii)	Any two from: • parent homozygous recessive/heterozygous / carriers (1) • one (recessive) allele inherited from each parent (1) • reference to homozygous recessive disorder/caused by recessive alleles (1)		2
	(iii)	• protein/CFTR is not modified/folded/protein has an incorrect shape(1) • protein/CFTR not transported (to its destination)(1)		2
	(iv)	1408 x 3 (1) 4224	Full marks for correct final answer. Max 2 marks ecf for final answer if it is correct from calculation	2
	(v)	• (deoxyribose) sugar(1) • joined to a (nitrogenous) base/named base(1) • phosphate group attached to sugar (1) 	Allow one mark only if all 3 components just listed Fully labelled diagram for full marks	3
(b)		any three from • deletion(1)	allow description of deletion	3

Type MS · Name the parts of a nucleotide and the DNA backbone

Answer 2 of 7 · Question 8 part a.v · 3 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a.v in this mark scheme extract

	• of one codon/TTT (1) • phenylalanine missing from protein (1) • shape of final protein changed/different protein formed (1)		

Total for Question 8 = 14 Marks

Type MS · Name the parts of a nucleotide and the DNA backbone

Answer 3 of 7 · Question 1 part a.i · 1 mark

2022 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

1 (a) (i)	**D** sugars A/B/C are incorrect as amino acids/bases/protein do not form part of backbone		1
(ii)	**B** bases A is incorrect as the structures labelled Y are bases C is incorrect as proteins are not incorporated into DNA structure D is incorrect as sugar is found in the backbone		1
(iii)	Any two from the following: • hydrogen bonds; • between complementary bases;	Allow description of complementary base pairing	2
(iv)	nucleus;		1
(b) (i)	Any two from the following: • RNA contains ribose sugar and DNA contains deoxyribose; • RNA is single-stranded and DNA is double stranded; • RNA contains uracil and DNA contains thymine;		2
(ii)	**D** mRNA and tRNA A/B/C are incorrect as pRNA/dRNA/dRNA are not types of RNA		1
(c)	**R** phosphate; **S** sugar/deoxyribose; **T** base;	Allow a named base	3
			Total 11

Type MS · Name the parts of a nucleotide and the DNA backbone

Answer 3 of 7 · Question 1 part a.i · 1 mark

2022 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name the parts of a nucleotide and the DNA backbone

Answer 4 of 7 · Question 1 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

1 (a) (i)	**D** sugars A/B/C are incorrect as amino acids/bases/protein do not form part of backbone		1
(ii)	**B** bases A is incorrect as the structures labelled Y are bases C is incorrect as proteins are not incorporated into DNA structure D is incorrect as sugar is found in the backbone		1
(iii)	Any two from the following: • hydrogen bonds; • between complementary bases;	Allow description of complementary base pairing	2
(iv)	nucleus;		1
(b) (i)	Any two from the following: • RNA contains ribose sugar and DNA contains deoxyribose; • RNA is single-stranded and DNA is double stranded; • RNA contains uracil and DNA contains thymine;		2
(ii)	**D** mRNA and tRNA A/B/C are incorrect as pRNA/dRNA/dRNA are not types of RNA		1
(c)	**R** phosphate; **S** sugar/deoxyribose; **T** base;	Allow a named base	3
			Total 11

Type MS · Name the parts of a nucleotide and the DNA backbone

Answer 4 of 7 · Question 1 part a.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name the parts of a nucleotide and the DNA backbone

Answer 5 of 7 · Question 1 part b.i · 2 marks

2023 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

Number				
1 (a) (i)	<table><tr><th rowspan="2">cell structure</th><th colspan="3">Function</th></tr><tr><th>makes proteins</th><th>contains DNA</th><th>allows substances in and out of cells</th></tr><tr><td>nucleus</td><td></td><td>✓</td><td></td></tr><tr><td>ribosomes</td><td>✓</td><td></td><td></td></tr><tr><td>cell membrane</td><td></td><td></td><td>✓</td></tr></table>	One mark for each correct row	3
(ii)	cell membrane/cell wall/cytoplasm/plasmid/DNA/ribosomes/flagellum;		1
(b) (i)	Part W phosphate (group); Part X sugar/ deoxyribose sugar;		2
(ii)	• complementary base pairing; • A-T, G-C; • hydrogen bonds between bases;		3
(c)	153 ÷ 3; 51;	Full marks for correct final answer Ecf allow correct final answer to calculation for 1 mark	2

Total for Question 1 = 11 Marks

Type MS · Name the parts of a nucleotide and the DNA backbone

Answer 6 of 7 · Question 2 part a.iii · 1 mark

2024 JUN · Paper 02 · find part a.iii in this mark scheme extract

Question number				
2	(a) (i)	• nucleus; • cytoplasm/RER/ribosomes;	Allow mitochondrion as an alternative mp	(2)
	(ii)	Any two from: • RNA is single stranded/DNA double stranded; • RNA contains uracil/DNA contains thymine ; • RNA contains ribose /DNA contains deoxyribose;	Allow DNA is double helix Do not allow thiamine	(2)
	(iii)	nucléotide ;	Allow mononucleotides	(1)
	(b)	Any three from • incorrect base/ nucleotide/codon sequence; • incorrect amino acid sequence/amino acid; • faulty/different/incorrect protein made; • reference to protein being a different shape/has a different function;	Allow named mutation/substitution/ deletion/addition/ insertion/eq	(3)

Total for Question 2 = 8 marks

Type MS · Name the parts of a nucleotide and the DNA backbone

Answer 7 of 7 · Question 2 part a.i · 2 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

2	(a)	(i)	X = (pentose) sugar/deoxyribose; Y = phosphate;	reject ribose reject phosphorus	2
		(ii)	DNA/deoxyribonucleic acid;		1
		(iii)	• double/two stranded; • thymine present; • only found in DNA (not RNA);		3
		(iv)	nucleus;		1
	(b)	(i)	hydrogen bonds;		1
		(ii)	• connect/hold/join, two strands/bases, together; • between complementary bases (pairs)/nucleotides; • adenine thymine/guanine cytosine;	ignore just A T G C	3
					11 marks

Type MS · Name the parts of a nucleotide and the DNA backbone

Answer 7 of 7 · Question 2 part a.i · 2 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks